Alcohols, Phenols and Ethers

Standard: National Eligibility cum Entrance Test (NEET UG) | **Subject:** Chemistry (Organic Chemistry)

NEET UG Strategy & Weightage Blueprint

The unit *Alcohols, Phenols and Ethers* is one of the most heavily weighted pillars of organic chemistry in NEET UG, historically contributing **2 to 3 questions (8 to 12 marks)** every year. The assessment profile focuses sharply on:

- **Mechanistic Reagents:** Acid-catalysed hydration vs. Hydroboration-Oxidation (anti-Markovnikov syn-addition), Oxymercuration-Demercuration, and Grignard additions to carbonyl compounds.
- **Acidity & Electronic Effects:** Resonance stability of the phenoxide ion, substituent effects (electron-withdrawing $-\mathrm{NO_2}$ vs. electron-donating $-\mathrm{OCH_3}$ / $-\mathrm{CH_3}$), and comparative pK_a trends.
- **Named Chemical Conversions:** Cumene process for phenol, Reimer-Tiemann and Kolbe reactions, Dow process, and Williamson ether synthesis (substrate elimination vs. substitution).
- **Regioselective Cleavage:** Unsymmetric ether cleavage by cold vs. hot concentrated HI ($\mathrm{S_N2}$ on $1^\circ/2^\circ$ alkyl groups vs. $\mathrm{S_N1}$ via stable 3° carbocations).

1. Learning Objectives

By studying this comprehensive module, aspirants will achieve the mastery required to solve NEET UG single-correct MCQs, assertion-reason items, and multi-statement matching problems on:

- Assigning IUPAC nomenclature to complex, polyhydric, and substituted alcohols, phenols, and ethers.
- Evaluating the bond angles, dipole moments, and hybridisation of oxygen in methanol (108.9°), phenol (109°), and methoxymethane (111.7°).
- Predicting the major and minor products of alcohol syntheses from alkenes, carbonyls, esters, and Grignard reagents.
- Applying the mechanisms of industrial phenol production (cumene hydroperoxide process) and laboratory preparations from diazonium salts.
- Differentiating primary, secondary, and tertiary alcohols using the Lucas reagent ($\mathrm{ZnCl_2} + \text{conc. } \mathrm{HCl}$) and catalytic dehydrogenation over Cu at $573\ \mathrm{K}$.
- Formulating electrophilic aromatic substitutions on phenol and anisole with precise ortho/para regioselectivity.

2. Classification & IUPAC Nomenclature

2.1 Structural Classification

Organic hydroxy and alkoxy derivatives are classified according to the number of functional groups and the electronic environment of the carbon atom bearing the oxygen atom:

- **Monohydric, Dihydric, Trihydric, and Polyhydric Alcohols:** Based on whether they contain one, two, three, or many hydroxyl ($-\mathrm{OH}$) groups per molecule. Examples: Ethanol (monohydric), Ethane-1,2-diol (glycol, dihydric), Propane-1,2,3-triol (glycerol, trihydric).
- **Primary (1°), Secondary (2°), and Tertiary (3°) Alcohols:** Based on the hybridization and degree of substitution of the carbon atom bound to the $-\mathrm{OH}$ group ($-\mathrm{CH_2OH}$, $>\mathrm{CHOH}$, or $-\mathrm{C(OH)}<$).
- **Allylic Alcohols:** Hydroxyl group attached to an sp^3-hybridized carbon adjacent to a carbon-carbon double bond: $\mathrm{CH_2=CH-CH_2OH}$.
- **Benzylic Alcohols:** Hydroxyl group attached to an sp^3-hybridized carbon atom adjacent to an aromatic ring: $\mathrm{C_6H_5-CH_2OH}$.
- **Vinylic Alcohols & Phenols:** Hydroxyl group bound directly to an sp^2-hybridized carbon. In vinylic alcohols, it is attached to an alkene carbon ($\mathrm{CH_2=CH-OH}$, tautomerizes to ethanal); in phenols, it is directly bound to an sp^2 aromatic carbon ($\mathrm{C_6H_5OH}$).
- **Ethers (Symmetrical vs. Unsymmetrical):** Classified as simple (symmetrical) if the two alkyl or aryl groups bound to oxygen are identical ($\mathrm{R=R'}$, e.g., diethyl ether $\mathrm{C_2H_5-O-C_2H_5}$), or mixed (unsymmetrical) if they differ ($\mathrm{R\neq R'}$, e.g., ethyl methyl ether $\mathrm{CH_3-O-C_2H_5}$).

2.2 IUPAC Rules and Systematic Names

In the IUPAC system, aliphatic alcohols are named by replacing the terminal *-e* of the parent alkane name with the suffix *-ol*. The longest carbon chain containing the $-\mathrm{OH}$ group is numbered from the end that gives the hydroxyl carbon the lowest locant. For polyhydric alcohols, the suffix *-diol*, *-triol*, etc., is appended to the full alkane name without dropping the terminal *-e* (e.g., ethane-1,2-diol). Ethers are named as *alkoxyalkanes*, where the larger alkyl group serves as the parent alkane, and the smaller alkyl group along with the oxygen is treated as an alkoxy substituent.

Table 1: Systematic IUPAC and Common Nomenclature of Selected Alcohols and Phenols

BOSONANCE

Chemical Formula	Common Name	IUPAC Name	Structural Class
$\mathrm{CH_3-CH_2-CH_2-OH}$	n-Propyl alcohol	Propan-1-ol	Primary (1°)
$\mathrm{CH_3-CH(OH)-CH_3}$	Isopropyl alcohol	Propan-2-ol	Secondary (2°)
$\mathrm{(CH_3)_3C-OH}$	tert-Butyl alcohol	2-Methylpropan-2-ol	Tertiary (3°)
$\mathrm{CH_2=CH-CH_2OH}$	Allyl alcohol	Prop-2-en-1-ol	Allylic (1°)
$\mathrm{HO-CH_2-CH_2-OH}$	Ethylene glycol	Ethane-1,2-diol	Dihydric ($1^\circ, 1^\circ$)
$\mathrm{HO-CH_2-CH(OH)-CH_2OH}$	Glycerol	Propane-1,2,3-triol	Trihydric
$\mathrm{C_6H_5-OH}$	Phenol (Carbolic acid)	Phenol	Monohydric Phenol
$\mathrm{o\text{-}CH_3-C_6H_4-OH}$	o-Cresol	2-Methylphenol	Substituted Phenol
$\mathrm{m\text{-}CH_3-C_6H_4-OH}$	m-Cresol	3-Methylphenol	Substituted Phenol
$\mathrm{p\text{-}CH_3-C_6H_4-OH}$	p-Cresol	4-Methylphenol	Substituted Phenol
$\mathrm{o\text{-}C_6H_4(OH)_2}$	Catechol	Benzene-1,2-diol	Dihydric Phenol
$\mathrm{m\text{-}C_6H_4(OH)_2}$	Resorcinol	Benzene-1,3-diol	Dihydric Phenol
$\mathrm{p\text{-}C_6H_4(OH)_2}$	Hydroquinone / Quinol	Benzene-1,4-diol	Dihydric Phenol

Table 2: Systematic IUPAC and Common Nomenclature of Selected Ethers

Chemical Formula	Common Name	IUPAC Name	Ether Type
$\mathrm{CH_3-O-CH_3}$	Dimethyl ether	Methoxymethane	Symmetrical
$\mathrm{C_2H_5-O-C_2H_5}$	Diethyl ether	Ethoxyethane	Symmetrical
$\mathrm{CH_3-O-CH_2CH_2CH_3}$	Methyl n-propyl ether	1-Methoxypropane	Unsymmetrical
$\mathrm{CH_3-O-CH(CH_3)_2}$	Methyl isopropyl ether	2-Methoxypropane	Unsymmetrical
$\mathrm{C_6H_5-O-CH_3}$	Anisole	Methoxybenzene	Aromatic (Alkyl aryl)
$\mathrm{C_6H_5-O-CH_2CH_3}$	Phenetole	Ethoxybenzene	Aromatic (Alkyl aryl)
$\mathrm{C_6H_5-O-(CH_2)_6CH_3}$	Phenyl n-heptyl ether	1-Phenoxyheptane	Aromatic (Alkyl aryl)

3. Structures of Functional Groups

The geometries around the oxygen atom in alcohols, phenols, and ethers reflect the interplay between hybridisation, non-bonding electron pair repulsions, and steric crowding:

- **Methanol ($\mathrm{CH_3OH}$):** The central oxygen atom is sp^3-hybridized with a tetrahedral electron-pair geometry. However, because lone pair-lone pair repulsions exceed bond pair-bond pair repulsions (VSEPR theory), the $\mathrm{C-O-H}$ bond angle is slightly compressed to 108.9° (less than the ideal tetrahedral angle of 109.5°). The $\mathrm{C-O}$ bond length is $142\ \mathrm{pm}$.
- **Phenol ($\mathrm{C_6H_5OH}$):** The oxygen atom is bonded to an sp^2-hybridized ring carbon. The $\mathrm{C-O-H}$ bond angle is 109°. Crucially, the $\mathrm{C-O}$ bond length in phenol is $136\ \mathrm{pm}$, significantly shorter than in methanol ($142\ \mathrm{pm}$). This bond contraction stems from two distinct factors:
  1. Partial double-bond character due to conjugation of the unshared electron pair on oxygen with the π-electron system of the aromatic ring.
  2. Greater electronegativity of the sp^2-hybridized aromatic carbon atom compared to the sp^3-hybridized carbon of methanol.
- **Methoxymethane ($\mathrm{CH_3OCH_3}$):** The oxygen atom is also sp^3-hybridized with two bonding pairs and two lone pairs. However, the $\mathrm{C-O-C}$ bond angle is widened to 111.7° (greater than 109.5°). This angle expansion is caused by strong repulsive steric interactions between the two bulky methyl groups. The $\mathrm{C-O}$ bond length is $141\ \mathrm{pm}$.

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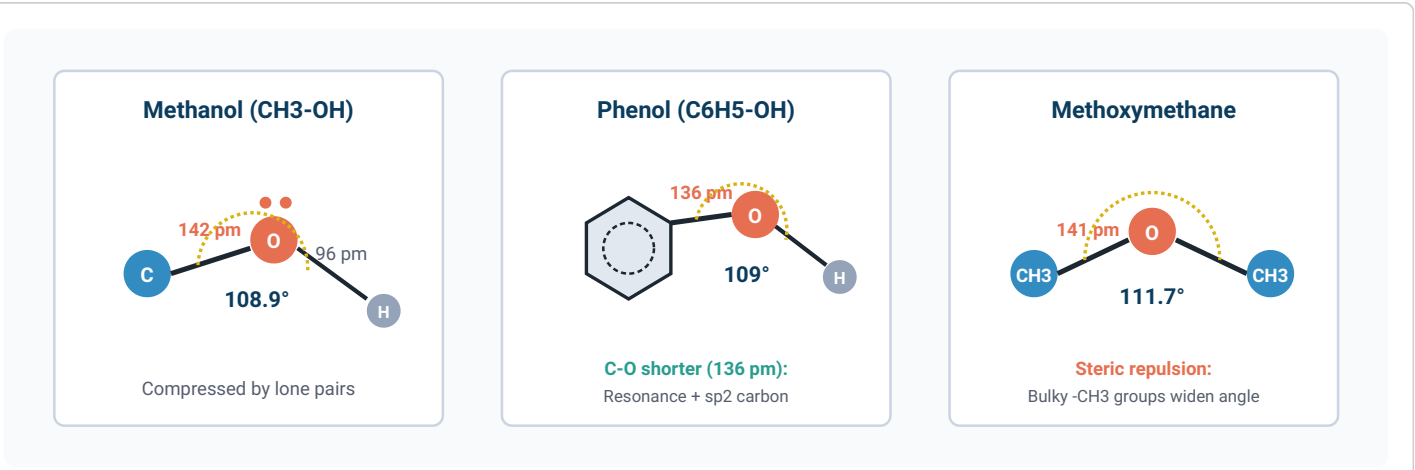


Diagram 1: *Structural parameters, bond angles, and bond lengths in methanol, phenol, and methoxymethane.*

Table 3: Quantitative Bond Parameter Comparison across Oxygen Functional Groups

Parameter	Methanol (CH_3OH)	Phenol (C_6H_5OH)	Methoxymethane (CH_3OCH_3)
Hybridization of Oxygen	sp^3	sp^3 (lone pair conjugated)	sp^3
Hybridization of Carbon attached to O	sp^3	sp^2	sp^3
$C-O$ **Bond Length**	$142\ \text{pm}$	$136\ \text{pm}$ (Partial double-bond)	$141\ \text{pm}$
$C-O-H$ **or** $C-O-C$ **Angle**	108.9° (Lone pair repulsion)	109°	111.7° (Steric alkyl crowding)

4. Synthesis of Alcohols and Phenols

4.1 Preparation of Alcohols

A. From Alkenes

1. **Acid-Catalyzed Hydration:** Alkenes react with water in the presence of dilute mineral acid (H_2SO_4) as a catalyst to yield alcohols according to **Markovnikov's rule**.

$$CH_3-CH=CH_2 + H_2O \xrightarrow{H^+} CH_3-CH(OH)-CH_3$$

NEET Mechanistic Check: The reaction proceeds via a planar carbocation intermediate. Consequently, skeletal carbocation rearrangements (1,2-hydride or 1,2-methyl shifts) readily occur if a more stable carbocation can form.

2. **Hydroboration-Oxidation:** Diborane (B_2H_6) reacts with alkenes to form trialkylboranes, which are subsequently oxidized by alkaline hydrogen peroxide (H_2O_2/OH^-) to yield alcohols.

$$6\,CH_3-CH=CH_2 + (BH_3)_2 \longrightarrow 2\,(CH_3-CH_2-CH_2)_3B \xrightarrow{3\,H_2O_2,\,OH^-} 6\,CH_3-CH_2-CH_2-OH + 2\,B(OH)_3$$

Key Features: Yields the pure **anti-Markovnikov** addition product. The net addition of H and OH occurs via concerted **syn-addition** (stereospecific) with **zero carbocation rearrangement**.

B. Reduction of Carbonyl Compounds and Carboxylic Acids

- **Aldehydes and Ketones:** Aldehydes are reduced to primary (1°) alcohols, while ketones are reduced to secondary (2°) alcohols using catalytic hydrogenation ($H_2/Ni, Pt, Pd$) or hydride transfer reagents like sodium borohydride ($NaBH_4$) in ethanol or lithium aluminium hydride ($LiAlH_4$) in ether.

$$R-CHO \xrightarrow{NaBH_4} R-CH_2OH$$
$$R-CO-R' \xrightarrow{NaBH_4} R-CH(OH)-R'$$

NEET Selectivity: $NaBH_4$ selectively reduces aldehydes and ketones without affecting carboxylic acids, esters, or isolated alkene double bonds.

- **Carboxylic Acids and Esters:** Carboxylic acids are reduced to primary alcohols by the powerful reducing agent $LiAlH_4$. Because $LiAlH_4$ is expensive, commercial synthesis converts acids to esters first, followed by catalytic hydrogenation:

$$R-COOH \xrightarrow{R'OH,\,H^+} R-COOR' \xrightarrow{H_2\ /\ \text{catalyst}} R-CH_2OH + R'OH$$

C. Addition of Grignard Reagents to Carbonyl Compounds

Nucleophilic addition of an alkyl/aryl magnesium halide ($RMgX$) to carbonyl groups followed by hydrolytic workup is a versatile route to synthesize all three classes of alcohols:

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- Methanal (Formaldehyde, HCHO) $\longrightarrow$ **Primary (1°) alcohol**.

$$\mathrm{HCHO} + \mathrm{RMgX} \longrightarrow \mathrm{R-CH_2-OMgX} \xrightarrow{\mathrm{H_2O}} \mathrm{R-CH_2OH} + \mathrm{Mg(OH)X}$$

- All other Aldehydes ($\mathrm{R'CHO}$) $\longrightarrow$ **Secondary (2°) alcohol**.

$$\mathrm{R'CHO} + \mathrm{RMgX} \longrightarrow \mathrm{R'-CH(R)-OMgX} \xrightarrow{\mathrm{H_2O}} \mathrm{R'-CH(OH)-R} + \mathrm{Mg(OH)X}$$

- Ketones ($\mathrm{R'COR''}$) $\longrightarrow$ **Tertiary (3°) alcohol**.

$$\mathrm{R'COR''} + \mathrm{RMgX} \longrightarrow \mathrm{R'R''C(R)-OMgX} \xrightarrow{\mathrm{H_2O}} \mathrm{R'R''(R)C-OH} + \mathrm{Mg(OH)X}$$

4.2 Preparation of Phenols

Phenols (carbolic acids) are manufactured industrially or synthesized in the laboratory via four primary pathways:

1. **From Haloarenes (Dow's Process):** Chlorobenzene is fused with molten sodium hydroxide at high temperature ($623\ \mathrm{K}$) and elevated pressure ($300\ \mathrm{atm}$) to produce sodium phenoxide, which upon acidification yields phenol.

$$\mathrm{C_6H_5Cl} + 2\,\mathrm{NaOH} \xrightarrow[300\ \mathrm{atm}]{623\ \mathrm{K}} \mathrm{C_6H_5ONa} + \mathrm{NaCl} + \mathrm{H_2O} \xrightarrow{\mathrm{H^+}} \mathrm{C_6H_5OH}$$

2. **From Benzenesulphonic Acid:** Benzene is sulphonated with oleum to give benzenesulphonic acid. Heating with molten NaOH forms sodium phenoxide, which yields phenol on acidification.

$$\mathrm{C_6H_6} \xrightarrow{\text{Oleum}} \mathrm{C_6H_5SO_3H} \xrightarrow{\text{(i) NaOH, (ii) } \mathrm{H^+}} \mathrm{C_6H_5OH}$$

3. **From Diazonium Salts:** Aniline is diazotized with nitrous acid ($\mathrm{NaNO_2 + HCl}$) at 273–$278\ \mathrm{K}$ to form benzenediazonium chloride. Warming with water or dilute acids liberates phenol and dinitrogen gas.

$$\mathrm{C_6H_5NH_2} \xrightarrow[273-278\ \mathrm{K}]{\mathrm{NaNO_2+HCl}} \mathrm{C_6H_5N_2^+Cl^-} \xrightarrow{\text{Warm } \mathrm{H_2O}} \mathrm{C_6H_5OH} + \mathrm{N_2}\uparrow + \mathrm{HCl}$$

4. **From Cumene (Isopropylbenzene) — Industrial Method:** Cumene is oxidized by aerial oxygen to **cumene hydroperoxide**. Subsequent cleavage with dilute aqueous acid converts it quantitatively to **phenol** and **propan-2-one (acetone)** as an economically valuable byproduct.

$$\mathrm{C_6H_5-CH(CH_3)_2} + \mathrm{O_2} \longrightarrow \mathrm{C_6H_5-C(CH_3)_2(OOH)} \xrightarrow{\mathrm{H^+,\ H_2O}} \mathrm{C_6H_5OH} + \mathrm{CH_3COCH_3}$$

5. Physical Properties & Intermolecular Forces

5.1 Boiling Points

The physical properties of alcohols and phenols are governed by the highly polar $\mathrm{O-H}$ bond, which enables extensive **intermolecular hydrogen bonding**:

- **Comparison with Hydrocarbons and Ethers:** Alcohols have significantly higher boiling points than alkanes, haloalkanes, and ethers of comparable molecular masses. For example, ethanol ($\mathrm{M_r} = 46$, bp $351\ \mathrm{K}$) boils far higher than methoxymethane ($\mathrm{M_r} = 46$, bp $248\ \mathrm{K}$) and propane ($\mathrm{M_r} = 44$, bp $231\ \mathrm{K}$).
- **Homologous Trends:** Within a homologous series, boiling points increase monotonically with carbon chain length due to larger surface area and stronger van der Waals dispersion forces.
- **Effect of Branching:** Among isomeric alcohols, boiling points decrease with increased branching ($1^\circ > 2^\circ > 3^\circ$). Branching causes the molecule to adopt a more compact spherical shape, decreasing surface area and weakening dispersion interactions:

$$\text{n-Butyl alcohol}\ (391\ \mathrm{K}) > \text{sec-Butyl alcohol}\ (373\ \mathrm{K}) > \text{tert-Butyl alcohol}\ (356\ \mathrm{K})$$

Table 4: Comparative Boiling Points of Isomeric Alcohols, Ethers, and Alkanes

Compound	Type	Molar Mass ($\mathrm{g\,mol^{-1}}$)	Boiling Point (K)
$\mathrm{CH_3CH_2CH_2CH_2OH}$	Primary Alcohol (Butan-1-ol)	74	391
$\mathrm{CH_3CH_2CH(OH)CH_3}$	Secondary Alcohol (Butan-2-ol)	74	373
$\mathrm{(CH_3)_3C-OH}$	Tertiary Alcohol (2-Methylpropan-2-ol)	74	356
$\mathrm{C_2H_5-O-C_2H_5}$	Ether (Ethoxyethane)	74	307.6
$\mathrm{CH_3(CH_2)_3CH_3}$	Alkane (n-Pentane)	72	309

5.2 Intramolecular vs. Intermolecular Hydrogen Bonding in Nitrophenols

A classic high-frequency NEET concept is the separation of ortho-nitrophenol and para-nitrophenol by steam distillation:

- **ortho-Nitrophenol:** The proximity of the $-\mathrm{OH}$ and $-\mathrm{NO_2}$ groups allows the formation of a 6-membered chelate ring via **intramolecular hydrogen bonding**. This prevents the hydroxyl proton from associating with neighbouring molecules, resulting in weaker intermolecular attractions, higher volatility, lower boiling point, and **steam volatility**.
- **para-Nitrophenol (and meta-nitrophenol):** The spatial separation of the two functional groups precludes intramolecular bonding. Instead, molecules associate strongly into polymeric chains via **intermolecular hydrogen bonding**, causing a higher boiling point, lower volatility, higher aqueous solubility, and complete lack of steam volatility.

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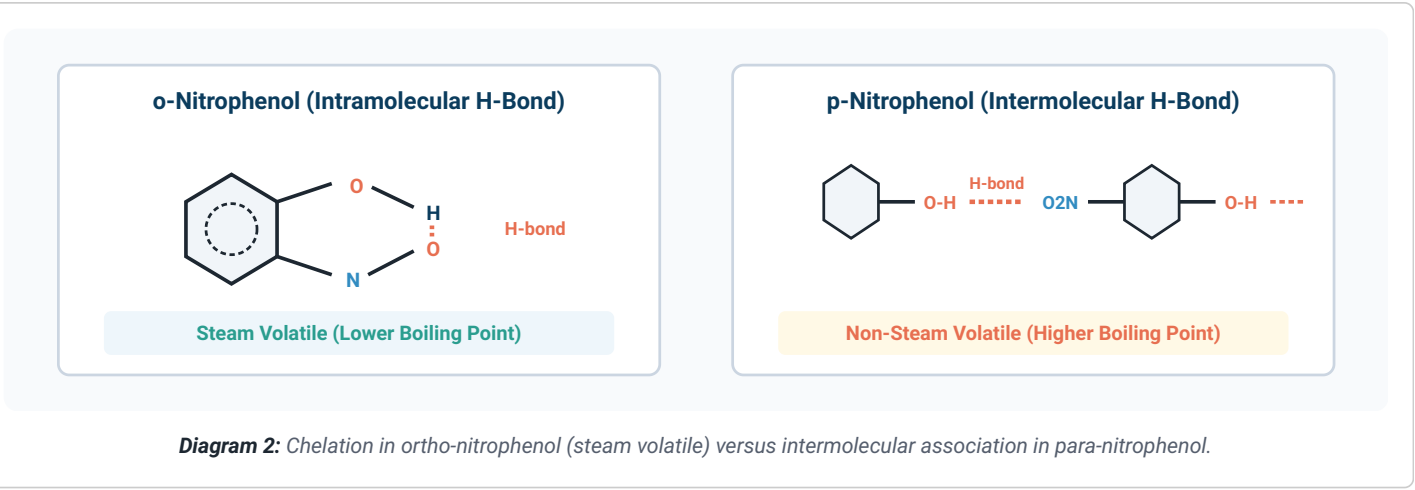


Diagram 2: *Chelation in ortho-nitrophenol (steam volatile) versus intermolecular association in para-nitrophenol.*

5.3 Solubility

Lower molecular mass alcohols and ethers form hydrogen bonds with water molecules, making them miscible with water. However, solubility decreases sharply as the size of the non-polar, hydrophobic alkyl/aryl group increases. High-mass alcohols and diaryl ethers are practically insoluble in water.

6. Chemical Reactions of Alcohols & Phenols

Alcohols act as versatile species: they react as **nucleophiles** when the $\mathrm{O-H}$ bond breaks, and as **electrophiles** when protonated, leading to $\mathrm{C-O}$ bond cleavage.

6.1 Reactions Involving Cleavage of the $\mathrm{O-H}$ Bond

A. Acidity of Alcohols and Phenols

1. **Reaction with Active Metals:** Alcohols and phenols react with electropositive metals like $\mathrm{Na, K, Al}$ to liberate dihydrogen gas:

$$2\,\mathrm{R-OH} + 2\,\mathrm{Na} \longrightarrow 2\,\mathrm{R-O^-Na^+} + \mathrm{H_2}\uparrow$$
$$6\,(\mathrm{CH_3})_3\mathrm{C-OH} + 2\,\mathrm{Al} \longrightarrow 2\,((\mathrm{CH_3})_3\mathrm{C-O})_3\mathrm{Al} + 3\,\mathrm{H_2}\uparrow$$

   Alcohols act as Brønsted acids by donating a proton to active metals. However, alcohols are weaker acids than water because electron-releasing alkyl groups ($+I$ effect) intensify electron density on oxygen, destabilizing the alkoxide anion. Acidity decreases in the order:

$$\mathbf{H_2O > 1^\circ\ Alcohol > 2^\circ\ Alcohol > 3^\circ\ Alcohol}$$

2. **Acidity of Phenols:** Phenols are markedly more acidic than aliphatic alcohols and react with aqueous sodium hydroxide (NaOH) to form sodium phenoxide:

$$\mathrm{C_6H_5OH} + \mathrm{NaOH} \rightleftharpoons \mathrm{C_6H_5ONa} + \mathrm{H_2O}$$

   Why are phenols more acidic than alcohols?
   - The $-\mathrm{OH}$ group is attached to an sp^2-hybridized carbon with higher electronegativity than an sp^3 carbon.
   - Deprotonation yields the **phenoxide ion**, which is resonance-stabilized by delocalization of the negative charge over the ortho and para positions of the benzene ring. Although neutral phenol also exhibits resonance, it involves unfavorable charge separation ($\mathrm{O^+}$ and $\mathrm{C^-}$), making the un-ionized form less stable than the phenoxide ion. In contrast, in alkoxide ions ($\mathrm{R-O^-}$), negative charge is localized strictly on oxygen with no resonance stabilization.

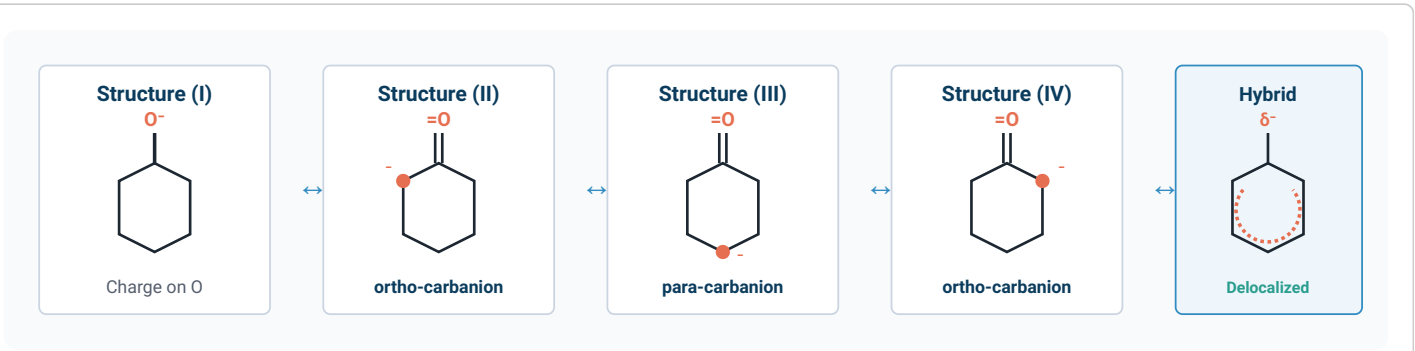


Diagram 3: *Resonance canonical structures of the phenoxide anion showing negative charge delocalization onto ortho and para positions.*

Table 5: Comparative Acid Dissociation Constants (pK_a) at $298\,\mathrm{K}$

Compound	Formula	pK_a Value	Acidity Assessment
Picric acid (2,4,6-Trinitrophenol)	$(\mathrm{NO_2})_3\mathrm{C_6H_2OH}$	**0.78**	Stronger than carboxylic acids

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Compound	Formula	pK_a Value	Acidity Assessment
o-Nitrophenol	$\text{o-NO}_2 - \text{C}_6\text{H}_4\text{OH}$	**7.2**	Strong ($-I, -R$ at ortho)
p-Nitrophenol	$\text{p-NO}_2 - \text{C}_6\text{H}_4\text{OH}$	**7.1**	Strong ($-I, -R$ at para; no chelate)
m-Nitrophenol	$\text{m-NO}_2 - \text{C}_6\text{H}_4\text{OH}$	**8.3**	Moderate ($-I$ effect only)
Phenol	$\text{C}_6\text{H}_5\text{OH}$	**10.0**	Reference aromatic hydroxy compound
m-Cresol	$\text{m-CH}_3 - \text{C}_6\text{H}_4\text{OH}$	**10.1**	Weakened by $+I$
p-Cresol	$\text{p-CH}_3 - \text{C}_6\text{H}_4\text{OH}$	**10.2**	Weakened by hyperconjugation / $+I$
o-Cresol	$\text{o-CH}_3 - \text{C}_6\text{H}_4\text{OH}$	**10.3**	Weakest among cresols
Ethanol	$\text{C}_2\text{H}_5\text{OH}$	**15.9**	Weaker acid than water ($pK_a = 15.7$)

B. Esterification

Alcohols and phenols react with carboxylic acids, acid anhydrides, and acid chlorides to form esters:

- With Carboxylic Acids (Fischer Esterification): Catalyzed by concentrated H_2SO_4. The reaction is reversible, and water is removed concurrently to drive equilibrium.

$$\text{Ar/R} - \text{OH} + \text{R}' - \text{COOH} \xrightleftharpoons{\text{H}^+} \text{Ar/R} - \text{OCO} - \text{R}' + \text{H}_2\text{O}$$

- With Acid Chlorides: Carried out in the presence of a base (such as **pyridine**) to neutralize the generated hydrochloric acid and shift equilibrium rightward.

$$\text{R} - \text{OH} + \text{R}'\text{COCl} \xrightarrow{\text{Pyridine}} \text{R} - \text{OCOR}' + \text{HCl}$$

- **Synthesis of Aspirin (Acetylsalicylic Acid):** Acetylation of salicylic acid (2-hydroxybenzoic acid) with acetic anhydride in the presence of trace acid yields aspirin, an essential analgesic and antipyretic:

$$\text{2-(HO)C}_6\text{H}_4\text{COOH} + (\text{CH}_3\text{CO})_2\text{O} \xrightarrow{\text{H}^+} \text{2-(CH_3COO)C}_6\text{H}_4\text{COOH} + \text{CH}_3\text{COOH}$$

6.2 Reactions Involving Cleavage of the Carbon-Oxygen ($\text{C} - \text{O}$) Bond

A. Reaction with Hydrogen Halides (Lucas Test)

Alcohols react with hydrogen halides to form alkyl halides: $\text{R} - \text{OH} + \text{HX} \xrightarrow{\text{ZnCl}_2} \text{R} - \text{X} + \text{H}_2\text{O}$. This forms the basis of the **Lucas Test** (equimolar concentrated HCl and anhydrous ZnCl_2) to distinguish $1^\circ, 2^\circ$, and 3° alcohols based on the rate of appearance of insoluble alkyl chloride turbidity:

- **Tertiary (3°) Alcohols:** React immediately at room temperature to form instantaneous cloudiness/turbidity (proceeds via stable 3° carbocation).
- **Secondary (2°) Alcohols:** Form turbidity within **5 to 10 minutes** at room temperature.
- **Primary (1°) Alcohols:** Do not produce turbidity at room temperature; solution remains clear until heated.

B. Reaction with Phosphorus Halides and Thionyl Chloride

- $3\,\text{R} - \text{OH} + \text{PCl}_3 \longrightarrow 3\,\text{R} - \text{Cl} + \text{H}_3\text{PO}_3$ (phosphorous acid).
- $\text{R} - \text{OH} + \text{PCl}_5 \longrightarrow \text{R} - \text{Cl} + \text{POCl}_3 + \text{HCl}\uparrow$.
- $\text{R} - \text{OH} + \text{SOCl}_2 \xrightarrow{\text{Pyridine}} \text{R} - \text{Cl} + \text{SO}_2\uparrow + \text{HCl}\uparrow$. This method gives the highest purity because both coproducts (SO_2 and HCl) are gases that escape automatically.

C. Acid-Catalyzed Dehydration of Alcohols

Alcohols undergo dehydration (loss of a water molecule) upon heating with concentrated protic acids like H_2SO_4 or H_3PO_4 to produce alkenes. The ease of dehydration follows the carbocation stability order: $3^\circ > 2^\circ > 1^\circ$.

- **Ethanol (1°):** Requires harsh conditions — conc. H_2SO_4 at $443\,\text{K}$.

$$\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[443\ \text{K}]{\text{conc. H}_2\text{SO}_4} \text{CH}_2 = \text{CH}_2 + \text{H}_2\text{O}$$

- **Propan-2-ol (2°):** Dehydrates under milder conditions — $85\%\ \text{H}_3\text{PO}_4$ at $440\,\text{K}$.

$$\text{CH}_3 - \text{CH(OH)} - \text{CH}_3 \xrightarrow[440\ \text{K}]{85\%\ \text{H}_3\text{PO}_4} \text{CH}_3 - \text{CH} = \text{CH}_2 + \text{H}_2\text{O}$$

- **2-Methylpropan-2-ol (3°):** Dehydrates under extremely mild conditions — $20\%\ \text{H}_3\text{PO}_4$ at $358\,\text{K}$.

$$(\text{CH}_3)_3\text{C} - \text{OH} \xrightarrow[358\ \text{K}]{20\%\ \text{H}_3\text{PO}_4} (\text{CH}_3)_2\text{C} = \text{CH}_2 + \text{H}_2\text{O}$$

BOSONANCE

Three-Step Mechanism of Acid-Catalyzed Dehydration of Ethanol to Ethene

1. **Step 1: Formation of protonated alcohol (Oxonium ion):** Rapid, reversible proton transfer from acid to the unshared electron pair on oxygen:

$$\mathrm{CH_3CH_2-OH + H^+ \xrightleftharpoons{fast} CH_3CH_2-\overset{+}{O}H_2}$$

2. **Step 2: Formation of carbocation:** Slowest, rate-determining step (RDS). Cleavage of the $\mathrm{C-O}$ bond releases water:

$$\mathrm{CH_3CH_2-\overset{+}{O}H_2 \xrightleftharpoons{slow\ (RDS)} CH_3-\overset{+}{C}H_2 + H_2O}$$

3. **Step 3: Elimination of a proton to form ethene:** Fast base attack by water on the β-hydrogen:

$$\mathrm{CH_3-\overset{+}{C}H_2 \xrightarrow{fast} CH_2=CH_2 + H^+}$$

D. Oxidation and Catalytic Dehydrogenation over Heated Copper

Oxidation of alcohols involves the cleavage of both an $\mathrm{O-H}$ and a $\mathrm{C-H}$ bond (dehydrogenation):

- **Primary (1°) Alcohols:** Oxidize to aldehydes using mild reagents such as **Pyridinium chlorochromate (PCC)** or Collins reagent in anhydrous $\mathrm{CH_2Cl_2}$. Strong oxidizing agents like acidified potassium permanganate ($\mathrm{KMnO_4}$) or Jones reagent ($\mathrm{CrO_3/H_2SO_4}$) oxidize primary alcohols directly to carboxylic acids:

$$\mathrm{R-CH_2OH \xrightarrow{PCC} R-CHO \xrightarrow{KMnO_4,\ H^+} R-COOH}$$

- **Secondary (2°) Alcohols:** Oxidize smoothly to ketones with chromic anhydride ($\mathrm{CrO_3}$) or potassium dichromate:

$$\mathrm{R-CH(OH)-R' \xrightarrow{CrO_3} R-CO-R'}$$

- **Tertiary (3°) Alcohols:** Do not possess an α-hydrogen and resist oxidation under neutral or alkaline conditions. Under vigorous acidic conditions ($\mathrm{HNO_3, KMnO_4}$), $\mathrm{C-C}$ bond cleavage occurs, yielding a mixture of carboxylic acids with fewer carbon atoms.
- **Dehydrogenation over Hot Copper (Cu at $573\ \mathrm{K}$):**
  - 1° Alcohol $\xrightarrow{\mathrm{Cu,\ 573\ K}}$ **Aldehyde** ($\mathrm{RCH_2OH \longrightarrow RCHO + H_2}$).
  - 2° Alcohol $\xrightarrow{\mathrm{Cu,\ 573\ K}}$ **Ketone** ($\mathrm{R_2CHOH \longrightarrow R_2CO + H_2}$).
  - 3° Alcohol $\xrightarrow{\mathrm{Cu,\ 573\ K}}$ **Alkene (Dehydration!)** Tert-butyl alcohol dehydrates to 2-methylpropene instead of dehydrogenating:

$$\mathrm{(CH_3)_3C-OH \xrightarrow{Cu,\ 573\ K} (CH_3)_2C=CH_2 + H_2O}$$

6.3 Distinctive Reactions of Phenol

Because the hydroxyl group strongly activates the aromatic ring by resonance ($+R$ effect) toward electrophilic aromatic substitution at the ortho and para positions, phenols exhibit unique reactivity:

1. **Electrophilic Halogenation:**
   - **Bromine in Non-polar Solvents ($\mathrm{Br_2}$ in $\mathrm{CS_2}$ or $\mathrm{CHCl_3}$ at $273\ \mathrm{K}$):** Monobromination occurs yielding predominantly **p-bromophenol** along with minor o-bromophenol. (No Lewis acid catalyst like $\mathrm{FeBr_3}$ is required because the $-\mathrm{OH}$ group strongly activates the ring).

$$\mathrm{C_6H_5OH + Br_2 \xrightarrow[273\ K]{CS_2} p\text{-}Br-C_6H_4OH\ (major) + o\text{-}Br-C_6H_4OH\ (minor)}$$

   - **Bromine Water ($\mathrm{Br_2/H_2O}$):** In aqueous media, phenol ionizes significantly to the phenoxide ion, which activates the ring so strongly that rapid trisubstitution occurs, giving a characteristic **white precipitate of 2,4,6-tribromophenol**:

$$\mathrm{C_6H_5OH + 3\,Br_2 \xrightarrow{H_2O} 2,4,6\text{-}Br_3C_6H_2OH\downarrow\ (white\ ppt) + 3\,HBr}$$

2. **Nitration:**
   - With **dilute $\mathrm{HNO_3}$ at $298\ \mathrm{K}$**: Yields a mixture of **ortho-nitrophenol** and **para-nitrophenol**, separable by steam distillation.
   - With **concentrated $\mathrm{HNO_3}$ in conc. $\mathrm{H_2SO_4}$**: Undergoes violent poly-nitration accompanied by substantial oxidative destruction to produce **2,4,6-trinitrophenol (Picric acid)** in low yield. Modern commercial preparation treats phenol with conc. $\mathrm{H_2SO_4}$ to form phenol-2,4-disulphonic acid first, followed by warming with conc. $\mathrm{HNO_3}$.

3. **Kolbe's Reaction:** Phenoxide ion, being more nucleophilic than neutral phenol, undergoes electrophilic substitution with the weak electrophile **carbon dioxide ($\mathrm{CO_2}$)** at $400\ \mathrm{K}$ and $4\text{–}7\ \mathrm{atm}$ pressure. Acidification produces **2-hydroxybenzoic acid (Salicylic acid)** as the major product.

$$\mathrm{C_6H_5ONa + CO_2 \xrightarrow[4-7\ atm]{400\ K} 2\text{-}(HO)C_6H_4COONa \xrightarrow{H^+} 2\text{-}(HO)C_6H_4COOH}$$

BOSONANCE

4. **Reimer-Tiemann Reaction:** Heating phenol with **chloroform** ($\mathrm{CHCl_3}$) in aqueous sodium hydroxide at $340\ \mathrm{K}$ introduces an aldehyde group ($-\mathrm{CHO}$) ortho to the hydroxyl group. Hydrolysis and acidification yield **Salicylaldehyde (2-hydroxybenzaldehyde)**:
$$\mathrm{C_6H_5OH + CHCl_3 + 3\,NaOH \longrightarrow 2\text{-}(HO)C_6H_4CHO + 3\,NaCl + 2\,H_2O}$$
NEET Mechanistic Check: The reactive electrophile is the neutral, divalent, electron-deficient intermediate **dichlorocarbene (** $:\mathrm{CCl_2}$), generated by base-induced α-elimination of HCl from chloroform. If carbon tetrachloride ($\mathrm{CCl_4}$) is substituted for chloroform, salicylic acid is formed instead.
5. **Reaction with Zinc Dust:** Heating phenol with zinc dust strips the oxygen atom, reducing phenol to **benzene**:
$$\mathrm{C_6H_5OH + Zn \xrightarrow{\Delta} C_6H_6 + ZnO}$$
6. **Oxidation with Chromic Acid:** Oxidation of phenol with acidified sodium dichromate ($\mathrm{Na_2Cr_2O_7/H_2SO_4}$) forms a conjugated cyclic diketone, **benzo-1,4-quinone (p-benzoquinone)**:
$$\mathrm{C_6H_5OH \xrightarrow{Na_2Cr_2O_7/H_2SO_4} O=C_6H_4=O\ (p\text{-benzoquinone})}$$
Phenols slowly oxidize in air to form complex quinone mixtures, turning commercial phenol samples pinkish-red over time.

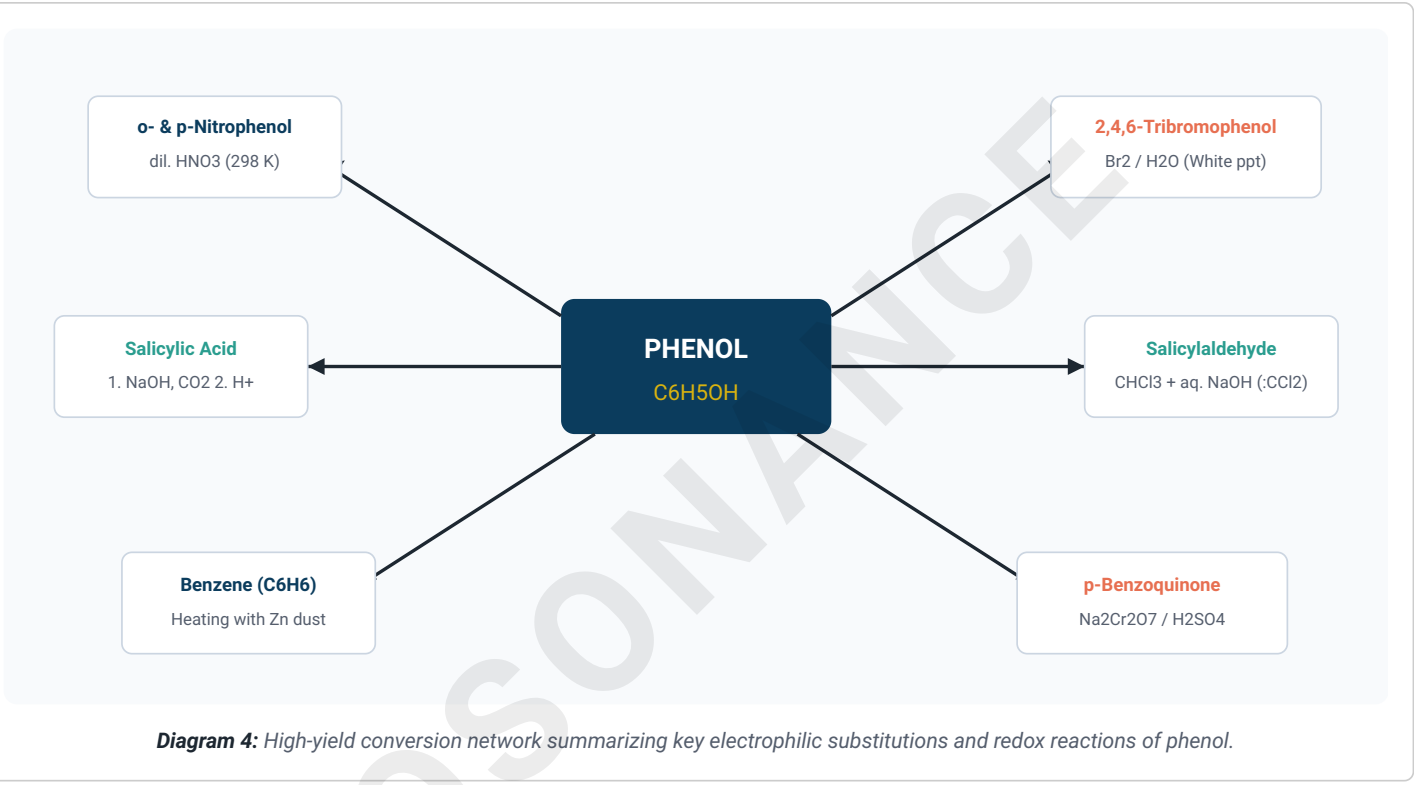


Diagram 4: *High-yield conversion network summarizing key electrophilic substitutions and redox reactions of phenol.*

7. Chemistry of Ethers

7.1 Preparation of Ethers

A. Industrial Dehydration of Alcohols

Ethanol undergoes intermolecular dehydration in the presence of concentrated sulfuric acid at $413\ \mathrm{K}$ to yield ethoxyethane via a bimolecular nucleophilic substitution ($\mathrm{S_N2}$) mechanism:
$$\mathrm{2\,CH_3CH_2OH \xrightarrow[413\ K]{conc.\ H_2SO_4} CH_3CH_2 - O - CH_2CH_3 + H_2O}$$

NEET Critical Temperature Warning: $413\ \mathrm{K}$ vs. $443\ \mathrm{K}$

- At $413\ \mathrm{K}$: Intermolecular nucleophilic attack of unprotonated alcohol on protonated alcohol dominates, producing **ether** ($\mathrm{S_N2}$).
- At $443\ \mathrm{K}$: Intramolecular β-elimination dominates, producing **ethene** ($\mathrm{E1}$).
- This synthetic method is strictly applicable to primary (1°) unhindered alcohols. Secondary (2°) and tertiary (3°) alcohols yield predominantly alkenes due to steric hindrance and competitive elimination.

BOSONANCE

B. Williamson Ether Synthesis

Williamson synthesis is the most reliable laboratory method to prepare symmetrical and unsymmetrical ethers. An alkyl halide reacts with a sodium alkoxide or aryloxide:

$$\mathrm{R-X + R'-O^-Na^+ \longrightarrow R'-O-R + NaX}$$

Key NEET Reaction Rules:

- The reaction proceeds via an $\mathrm{S_N2}$ **substitution** pathway. Therefore, the alkyl halide ($\mathrm{R-X}$) must be **primary** (1°) and sterically unhindered.
- Sodium alkoxides are strong nucleophiles and powerful Brønsted bases. If a **secondary** (2°) **or tertiary** (3°) **alkyl halide** is used, elimination dominates completely over substitution, producing an **alkene** as the exclusive major product (E2 mechanism).
- **To prepare tert-butyl methyl ether:**
  - *Correct Path:* Sodium tert-butoxide + Chloromethane/Bromomethane

$$\mathrm{(CH_3)_3C-O^-Na^+ + CH_3-Br \xrightarrow{S_N2} (CH_3)_3C-O-CH_3 + NaBr}$$

  - *Incorrect Path:* Sodium methoxide + tert-Butyl bromide

$$\mathrm{CH_3O^-Na^+ + (CH_3)_3C-Br \xrightarrow{E2} CH_2=C(CH_3)_2\ (2\text{-methylpropene}) + CH_3OH + NaBr}$$

- Aryl halides and vinyl halides cannot be used as the halide component because partial double-bond character makes the $\mathrm{C-X}$ bond unreactive toward nucleophilic substitution. Phenolic ethers (e.g., anisole) are prepared by reacting sodium phenoxide with methyl halides:

$$\mathrm{C_6H_5O^-Na^+ + CH_3-I \longrightarrow C_6H_5-O-CH_3 + NaI}$$

7.2 Chemical Reactions of Ethers: Cleavage of the $\mathrm{C-O}$ Bond

Ethers are comparatively unreactive, but undergo $\mathrm{C-O}$ bond cleavage when heated with concentrated hydrogen halides ($\mathrm{HI > HBr > HCl}$):

$$\mathrm{R-O-R' + HX \longrightarrow R-X + R'-OH}$$

If excess hydrogen halide is used at elevated temperatures, the formed alcohol reacts further to yield a second equivalent of alkyl halide:

$$\mathrm{R'-OH + HX \longrightarrow R'-X + H_2O}$$

Regioselectivity and Mechanism Rules for Asymmetric Ethers

1. **Case 1: Both Alkyl Groups are Primary or Secondary (1° or 2°):** The reaction proceeds via an $\mathrm{S_N2}$ **mechanism**. After protonation of the ether oxygen, the iodide nucleophile ($\mathrm{I^-}$) selectively attacks the **less sterically hindered alkyl group**:
$$\mathrm{CH_3-O-CH_2CH_2CH_3 + HI \longrightarrow CH_3-I + CH_3CH_2CH_2-OH}$$
2. **Case 2: One Alkyl Group is Tertiary (3°):** The reaction shifts to an $\mathrm{S_N1}$ **mechanism**. After protonation, the $\mathrm{C-O}$ bond breaks to form a stable tertiary carbocation, which is then trapped by iodide. Hence, the **tertiary group forms the alkyl iodide**:
$$\mathrm{(CH_3)_3C-O-CH_3 + HI \longrightarrow (CH_3)_3C-I + CH_3-OH}$$
3. **Case 3: Alkyl Aryl Ethers (e.g., Anisole):** Protonation forms the oxonium ion $\mathrm{C_6H_5-\overset{+}{O}(H)-CH_3}$. The $\mathrm{sp^2\ C(aromatic)-O}$ bond possesses partial double-bond character from resonance and resists cleavage. Nucleophilic attack by iodide takes place at the sp^3 methyl carbon via $\mathrm{S_N2}$, yielding **phenol and methyl iodide**:
$$\mathrm{C_6H_5-O-CH_3 + HI \longrightarrow C_6H_5-OH + CH_3-I}$$
Phenol does not react further with HI to form iodobenzene because the aromatic carbon-oxygen bond cannot be cleaved under these conditions.

BOSONANCE

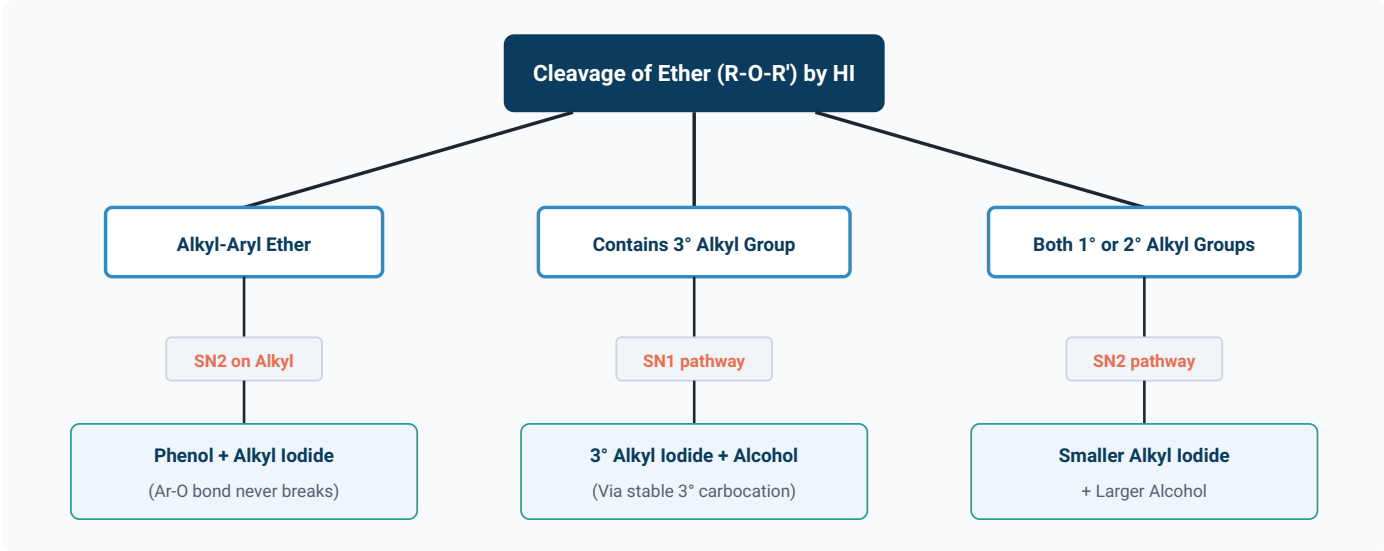


Diagram 5: *Algorithmic decision logic for predicting products and mechanisms of ether cleavage with* HI.

7.3 Electrophilic Aromatic Substitution in Anisole

In alkyl aryl ethers (such as anisole, $\mathrm{C_6H_5OCH_3}$), the methoxy group ($-\mathrm{OCH_3}$) is ortho- and para-directing and activates the benzene ring due to the $+R$ resonance effect of oxygen's lone pairs:

- **Halogenation:** Anisole undergoes bromination with bromine in ethanoic acid without any $\mathrm{FeBr_3}$ catalyst to give **90% para-bromoanisole** and minor ortho-bromoanisole.
- **Friedel-Crafts Reactions:**
  - *Alkylation:* Anisole reacts with chloromethane in the presence of anhydrous $\mathrm{AlCl_3}$ to give a mixture of 2-methoxytoluene (minor) and **4-methoxytoluene (major)**.
  - *Acylation:* Reaction with ethanoyl chloride ($\mathrm{CH_3COCl}$) and anhydrous $\mathrm{AlCl_3}$ yields 2-methoxyacetophenone (minor) and **4-methoxyacetophenone (major)**.
- **Nitration:** Anisole reacts with a mixture of concentrated sulfuric and nitric acids to yield a mixture of 2-nitroanisole (minor) and **4-nitroanisole (major)**.

8. Commercially Important Alcohols

8.1 Methanol ($\mathrm{CH_3OH}$, Wood Spirit)

- **Historical Origin:** Originally obtained by the destructive distillation of wood.
- **Modern Industrial Manufacture:** Catalytic hydrogenation of carbon monoxide in the presence of a zinc oxide-chromium oxide catalyst under high temperature and pressure:

$$\mathrm{CO + 2\,H_2 \xrightarrow[200-300\ atm,\ 573-673\ K]{ZnO-Cr_2O_3} CH_3OH}$$

- **Toxicity & Biological Hazard:** Methanol is highly poisonous. Ingestion of even small quantities (10–$30\ \mathrm{mL}$) can cause blindness, while larger doses cause death due to respiratory failure. In the liver, methanol is oxidized by the enzyme alcohol dehydrogenase to **methanal (formaldehyde)**, and subsequently to **methanoic acid (formic acid)**:

$$\mathrm{CH_3OH \xrightarrow{Alcohol\ dehydrogenase} HCHO \longrightarrow HCOOH}$$

  Formaldehyde coagulates cellular protoplasm (analogous to boiling an egg), while formic acid induces severe metabolic acidosis and destroys retinal cells, causing optic neuropathy and permanent blindness.
- **Clinical Antidote:** Intravenous infusion of dilute **ethanol**. Ethanol acts as a competitive substrate for alcohol dehydrogenase (having a much higher affinity for the enzyme), which halts the metabolic oxidation of methanol and allows it to be excreted harmlessly.

8.2 Ethanol ($\mathrm{C_2H_5OH}$)

- **Industrial Preparation by Fermentation:** Fermentation of sugars from molasses, sugarcane, or grains:

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \xrightarrow{Invertase} C_6H_{12}O_6\ (Glucose) + C_6H_{12}O_6\ (Fructose)}$$

$$\mathrm{C_6H_{12}O_6 \xrightarrow{Zymase} 2\,C_2H_5OH + 2\,CO_2\uparrow}$$

  Fermentation is strictly anaerobic. If air is admitted, bacterium *Acetobacter* oxidizes ethanol to acetic acid, spoiling the mixture. Fermentation self-terminates when ethanol concentration reaches 14% because higher alcohol levels denature zymase.
- **Denaturation of Alcohol:** Commercial alcohol is rendered unfit for drinking by adding poisonous compounds like **methanol** (5%) and foul-smelling basic pyridine, along with a blue dye like copper sulfate. This blend is known as **denatured spirit / methylated**

BOSONANCE

spirit.

9. Canonical Worked Examples (NEET UG Level)

Worked Example 1: Regiochemical Outcome of Alkene Hydration

MECHANISTIC STRATEGY

Problem: Predict the major organic product formed when 3,3-dimethylbut-1-ene undergoes: (a) Acid-catalyzed hydration ($\mathrm{H_2O/H_2SO_4}$), and (b) Hydroboration-oxidation ($\mathrm{B_2H_6}$ followed by $\mathrm{H_2O_2/OH^-}$).

Step-by-Step Solution:

Substrate: $\mathrm{(CH_3)_3C-CH=CH_2}$

(a) Acid-Catalyzed Hydration:

1. Electrophilic addition of $\mathrm{H^+}$ generates a secondary carbocation: $\mathrm{(CH_3)_3C-\overset{+}{C}H-CH_3}$ (2°).
2. A **1,2-methyl shift** occurs from the adjacent quaternary carbon to form a more stable tertiary carbocation:

$$\mathrm{(CH_3)_3C-\overset{+}{C}H-CH_3 \xrightarrow{1,2-shift} (CH_3)_2\overset{+}{C}-CH(CH_3)_2}\ (3^\circ)$$

3. Attack of $\mathrm{H_2O}$ and deprotonation yields **2,3-dimethylbutan-2-ol**:

$$\mathrm{(CH_3)_2C(OH)-CH(CH_3)_2}$$

(b) Hydroboration-Oxidation:

Proceeds via a cyclic 4-membered transition state with concerted addition of $\mathrm{-H}$ and $\mathrm{-BH_2}$. No carbocation is formed, precluding rearrangements. The reaction gives the pure **anti-Markovnikov** product: **3,3-dimethylbutan-1-ol**, $\mathrm{(CH_3)_3C-CH_2-CH_2OH}$.

Worked Example 2: Ether Cleavage Analysis

$\mathrm{S_N1}$ VS. $\mathrm{S_N2}$ LOGIC

Problem: Identify the products obtained when 1-methoxy-4-(2-methylpropan-2-yl)benzene is treated with one equivalent of concentrated HI at room temperature.

Step-by-Step Solution:

Structure: The molecule is $\mathrm{p\text{-}(CH_3)_3C-C_6H_4-O-CH_3}$ (an alkyl aryl ether).

1. Protonation occurs at the ether oxygen atom: $\mathrm{p\text{-}(CH_3)_3C-C_6H_4-\overset{+}{O}(H)-CH_3}$.
2. The $\mathrm{C(aromatic)-O}$ bond has partial double-bond character due to resonance with the benzene ring and cannot be cleaved.
3. The iodide nucleophile ($\mathrm{I^-}$) attacks the unhindered sp^3-hybridized methyl group via an $\mathrm{S_N2}$ pathway.
4. Products formed: **4-tert-butylphenol** ($\mathrm{p\text{-}(CH_3)_3C-C_6H_4-OH}$) and **iodomethane** ($\mathrm{CH_3I}$).

10. NEET UG Exam-Pattern Practice Questions

The following original practice questions adhere strictly to current National Testing Agency (NTA) NEET UG formats: Single-Correct MCQs, Assertion-Reason, Two-Statement (Statement I & II), and Matching items.

BOSONANCE

Practice Question 1

SINGLE-CORRECT MCQ

In the reaction sequence:

$$\text{Phenol} \xrightarrow{\text{Zn dust}} \text{X} \xrightarrow[\text{anhyd. } AlCl_3]{CH_3Cl} \text{Y} \xrightarrow{\text{alkaline } KMnO_4,\ \Delta} \text{Z}$$

The final compound Z is:

(A) Benzaldehyde

(B) Benzoic acid

(C) Benzophenone

(D) o-Cresol

Correct Answer: (B)

Step-by-step Explanation:

1. Reduction of phenol with zinc dust removes oxygen to yield benzene: $\text{X} = C_6H_6$.
2. Friedel-Crafts alkylation of benzene with CH_3Cl and anhydrous $AlCl_3$ yields toluene: $\text{Y} = C_6H_5CH_3$.
3. Vigorous oxidation of the benzylic methyl group of toluene with hot alkaline $KMnO_4$ followed by acidification yields benzoic acid: $\text{Z} = C_6H_5COOH$.

Practice Question 2

SINGLE-CORRECT MCQ

An organic compound A ($C_4H_{10}O$) does not react with sodium metal. When heated with excess concentrated hydroiodic acid (HI), it gives only one type of alkyl halide containing four carbon atoms. The structure of compound A is:

(A) Ethoxyethane

(B) 1-Methoxypropane

(C) 2-Methoxypropane

(D) 2-Methylpropan-2-ol

Correct Answer: (A)

Step-by-step Explanation:

1. Compound A does not react with active sodium metal, ruling out alcohols (eliminating option D). Thus, A is an ether of molecular formula $C_4H_{10}O$.
2. Treatment of an unsymmetrical ether like 1-methoxypropane or 2-methoxypropane with excess HI produces two distinct alkyl iodides: CH_3I and propyl/isopropyl iodide.
3. Treatment of symmetrical ethoxyethane ($C_2H_5 - O - C_2H_5$) with excess HI yields two moles of a single alkyl halide: ethyl iodide (C_2H_5I). Because the question specifies a symmetrical four-carbon ether that yields identical fragments, compound A is ethoxyethane.

BOSONANCE

Practice Question 3

SINGLE-CORRECT MCQ

Which of the following compounds exhibits the highest boiling point?

(A) Pentan-1-ol

(B) 2-Methylbutan-2-ol

(C) Ethoxypropane

(D) n-Pentane

Correct Answer: (A)

Step-by-step Explanation:

1. All four compounds possess comparable molecular weights ($\mathrm{M_r} \approx 72$–88).
2. Alkanes and ethers cannot participate in intermolecular hydrogen bonding with themselves, so n-pentane and ethoxypropane have significantly lower boiling points (~ 309–$337\,\mathrm{K}$).
3. Both pentan-1-ol and 2-methylbutan-2-ol form intermolecular hydrogen bonds. However, pentan-1-ol is an unbranched primary alcohol with a linear chain and larger surface area, leading to stronger van der Waals dispersion forces. 2-Methylbutan-2-ol is a branched tertiary alcohol with a compact spherical shape, which lowers its boiling point. Hence, pentan-1-ol has the highest boiling point ($411\,\mathrm{K}$).

Practice Question 4

ASSERTION-REASON

Assertion (A): Phenol reacts with bromine in carbon disulphide ($\mathrm{CS_2}$) at $273\,\mathrm{K}$ to give p-bromophenol as the major product without requiring a Lewis acid catalyst.

Reason (R): The hydroxyl group attached to the benzene ring possesses a strong $+R$ activating effect, making the aromatic nucleus electron-rich.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Step-by-step Explanation:

In ordinary benzene, halogenation requires a Lewis acid ($\mathrm{FeBr_3}$) to polarize bromine into $\mathrm{Br^+}$. In phenol, the unshared lone pair on oxygen participates in strong $+R$ resonance with the ring π-system, substantially increasing electron density at the ortho and para positions. As a result, the ring is activated enough to polarize and react with molecular bromine even in non-polar solvents like $\mathrm{CS_2}$ without any catalyst. Thus, both (A) and (R) are true and (R) correctly explains (A).

BOSONANCE

Practice Question 5

STATEMENT I AND STATEMENT II

Given below are two statements:

Statement I: The $\mathrm{C-O-H}$ bond angle in methanol is slightly less than the tetrahedral angle of 109.5°.

Statement II: The $\mathrm{C-O-C}$ bond angle in methoxymethane is smaller than the tetrahedral angle due to lone pair-lone pair repulsions on the oxygen atom.

In light of the above statements, choose the most appropriate answer:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C)

Step-by-step Explanation:

1. **Statement I is correct:** In methanol, the oxygen atom is sp^3-hybridized. Unshared lone pairs exert greater repulsion on bonding pairs, compressing the bond angle to 108.9° (less than 109.5°).
2. **Statement II is incorrect:** In methoxymethane, the $\mathrm{C-O-C}$ bond angle is 111.7°, which is *greater* than the tetrahedral angle. This expansion is driven by strong steric repulsive forces between the bulky methyl groups, which override lone pair-lone pair compression.

Practice Question 6

SINGLE-CORRECT MCQ

When tert-butyl ethyl ether is treated with one equivalent of concentrated hydroiodic acid (HI), the products formed are:

(A) tert-Butyl iodide and Ethanol

(B) tert-Butyl alcohol and Ethyl iodide

(C) tert-Butyl iodide and Ethyl iodide

(D) 2-Methylpropene and Ethanol

Correct Answer: (A)

Step-by-step Explanation:

1. Protonation of the oxygen atom yields the oxonium ion: $(\mathrm{CH_3})_3\mathrm{C-\overset{+}{O}(H)-C_2H_5}$.
2. Because one alkyl group is tertiary (3°), the reaction proceeds via an $\mathrm{S_N1}$ **mechanism**. The tertiary $\mathrm{C-O}$ bond undergoes heterolytic cleavage to form a stable tertiary butyl carbocation, $(\mathrm{CH_3})_3\mathrm{C^+}$, while releasing neutral ethanol ($\mathrm{C_2H_5OH}$).
3. The iodide ion ($\mathrm{I^-}$) rapidly attacks the carbocation to form tert-butyl iodide, $(\mathrm{CH_3})_3\mathrm{C-I}$.

BOSONANCE

Practice Question 7

SINGLE-CORRECT MCQ

The correct descending order of acidic strength among the following hydroxy compounds is:

(I) 4-Nitrophenol **(II)** Phenol **(III)** 4-Cresol **(IV)** 2,4,6-Trinitrophenol

(A) IV > I > II > III

(B) IV > II > I > III

(C) I > IV > II > III

(D) IV > I > III > II

Correct Answer: (A)

Step-by-step Explanation:

1. Picric acid (IV, 2,4,6-trinitrophenol) has three strong electron-withdrawing $-\mathrm{NO_2}$ groups ($-I, -R$), which disperse negative charge across the ring to give a pK_a of 0.78 (strongest acid).
2. 4-Nitrophenol (I) has one $-\mathrm{NO_2}$ group at the para position stabilizing the phenoxide ion through $-R$ and $-I$ effects ($pK_a = 7.1$).
3. Phenol (II) is unsubstituted ($pK_a = 10.0$).
4. 4-Cresol (III) contains an electron-releasing $-\mathrm{CH_3}$ group at the para position ($+I$ and hyperconjugation), which intensifies the negative charge and destabilizes the phenoxide ion ($pK_a = 10.2$).
5. Therefore, the correct acidity sequence is **IV > I > II > III**.

Practice Question 8

MATCHING LIST

Match **List-I** (Name of Chemical Transformation) with **List-II** (Characteristic Reagent or Reactive Intermediate):

List-I	List-II
(P) Reimer-Tiemann Reaction	(1) Cumene hydroperoxide rearrangement
(Q) Kolbe's Reaction	(2) Dichlorocarbene intermediate (: $\mathrm{CCl_2}$)
(R) Industrial synthesis of Phenol	(3) $\mathrm{CO_2}$ at $400\ \mathrm{K}$, $4–7\ \mathrm{atm}$
(S) Lucas Test for Alcohols	(4) Anhydrous $\mathrm{ZnCl_2} + \mathrm{conc.\ HCl}$

(A) P-2, Q-3, R-1, S-4

(B) P-3, Q-2, R-1, S-4

(C) P-2, Q-1, R-3, S-4

(D) P-4, Q-3, R-1, S-2

Correct Answer: (A)

Step-by-step Explanation:

- **(P) Reimer-Tiemann Reaction:** Employs chloroform and alkali; proceeds via the electrophilic *dichlorocarbene* intermediate (matches **2**).
- **(Q) Kolbe's Reaction:** Phenoxide reacts with carbon dioxide under pressure at $400\ \mathrm{K}$ (matches **3**).
- **(R) Industrial synthesis of Phenol:** Aerial oxidation of cumene generates cumene hydroperoxide, which cleaves in acid to phenol and acetone (matches **1**).
- **(S) Lucas Test:** Uses anhydrous $\mathrm{ZnCl_2}$ and concentrated HCl (matches **4**).

BOSONANCE

Practice Question 9

SINGLE-CORRECT MCQ

Which set of reactants produces tert-butyl ethyl ether in the highest yield via the Williamson synthesis?

(A) Sodium ethoxide + tert-Butyl bromide

(B) Sodium tert-butoxide + Bromoethane

(C) tert-Butyl alcohol + Ethanol in equal amounts

(D) Potassium tert-butoxide + Ethene

Correct Answer: (B)

Step-by-step Explanation:

1. In Williamson synthesis ($\mathrm{S_N2}$), the alkyl halide must be unhindered to permit backside nucleophilic attack.
2. In option (A), tert-butyl bromide is a tertiary halide, and sodium ethoxide is a strong base. Strong steric hindrance precludes substitution, leading to exclusive **E2 elimination** to yield 2-methylpropene.
3. In option (B), bromoethane is a primary alkyl halide, while potassium/sodium tert-butoxide acts as a bulky nucleophile. Nucleophilic displacement of bromide proceeds cleanly via $\mathrm{S_N2}$ to afford tert-butyl ethyl ether in high yield.

Practice Question 10

SINGLE-CORRECT MCQ

A secondary alcohol X on treatment with heated copper at $573\,\mathrm{K}$ gives compound Y. Compound Y reacts with iodine and aqueous sodium hydroxide to form a yellow precipitate with an antiseptic odour. Compound X can be:

(A) Pentan-3-ol

(B) Propan-1-ol

(C) Butan-2-ol

(D) 2-Methylpropan-2-ol

Correct Answer: (C)

Step-by-step Explanation:

1. Passage of a secondary alcohol over copper at $573\,\mathrm{K}$ causes catalytic dehydrogenation to form a ketone Y.
2. Compound Y gives a positive iodoform test (yellow precipitate of $\mathrm{CHI_3}$ with an antiseptic smell), which requires a **methyl ketone** grouping ($\mathrm{CH_3-CO-R}$).
3. Dehydrogenation of butan-2-ol (X) produces butan-2-one ($\mathrm{CH_3-CO-CH_2CH_3}$), which contains a methyl ketone group and readily forms iodoform.
4. Pentan-3-ol dehydrogenates to pentan-3-one ($\mathrm{CH_3CH_2-CO-CH_2CH_3}$), which is not a methyl ketone and tests negative for iodoform. Propan-1-ol is a primary alcohol, and 2-methylpropan-2-ol is a tertiary alcohol that undergoes dehydration rather than dehydrogenation.

BOSONANCE

Practice Question 11

ASSERTION-REASON

Assertion (A): ortho-Nitrophenol is steam volatile and has a lower boiling point than para-nitrophenol.

Reason (R): ortho-Nitrophenol exhibits intramolecular hydrogen bonding, whereas para-nitrophenol undergoes extensive intermolecular hydrogen bonding.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Step-by-step Explanation:

In ortho-nitrophenol, the hydroxyl proton forms an intramolecular hydrogen bond with the adjacent nitro oxygen, creating a 6-membered ring (chelation). This prevents the molecule from associating with neighbouring molecules, increasing volatility and lowering the boiling point. In contrast, para-nitrophenol associates strongly via intermolecular hydrogen bonding into extended polymeric networks, which suppresses volatility. Thus, both statements are true and Reason (R) correctly explains Assertion (A).

Practice Question 12

SINGLE-CORRECT MCQ

The principal organic byproduct formed in the industrial manufacture of phenol from cumene is:

(A) Acetaldehyde

(B) Acetophenone

(C) Propan-2-one (Acetone)

(D) Benzoic acid

Correct Answer: (C)

Step-by-step Explanation:

In the cumene process, cumene (isopropylbenzene) is auto-oxidized to cumene hydroperoxide. Subsequent treatment with dilute sulfuric acid triggers an acid-catalyzed rearrangement where the phenyl group migrates to oxygen, producing **phenol** and equimolar **propan-2-one (acetone)**. Industrial recovery of acetone makes this method economically advantageous.

BOSONANCE

Practice Question 13

STATEMENT I AND STATEMENT II

Given below are two statements:

Statement I: Aspirin is chemically 2-acetoxybenzoic acid and is prepared by the acetylation of salicylic acid with acetic anhydride.

Statement II: Salicylic acid is synthesized from sodium phenoxide and carbon dioxide via the Reimer-Tiemann reaction.

In light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C)

Step-by-step Explanation:

1. **Statement I is correct:** Acetylation of the phenolic $-\mathrm{OH}$ group of salicylic acid with acetic anhydride in the presence of trace acid yields acetylsalicylic acid (aspirin).
2. **Statement II is incorrect:** The reaction of sodium phenoxide with $\mathrm{CO_2}$ under pressure to produce salicylic acid is **Kolbe's reaction**. The Reimer-Tiemann reaction uses chloroform and aqueous NaOH to form *salicylaldehyde*.

Practice Question 14

SINGLE-CORRECT MCQ

Treatment of anisole with a mixture of concentrated $\mathrm{HNO_3}$ and concentrated $\mathrm{H_2SO_4}$ produces:

(A) 4-Nitroanisole as the major product

(B) 2-Nitroanisole as the major product

(C) 3-Nitroanisole as the major product

(D) 2,4,6-Trinitroanisole exclusively

Correct Answer: (A)

Step-by-step Explanation:

The methoxy group ($-\mathrm{OCH_3}$) in anisole is an activating, ortho- and para-directing group due to $+R$ resonance. Nitration yields a mixture of ortho- and para-nitroanisole. The **para-isomer (4-nitroanisole)** is the major product because the bulky methoxy group creates steric hindrance at the adjacent ortho positions.

Practice Question 15

SINGLE-CORRECT MCQ

An unknown alcohol reacts instantly with Lucas reagent at room temperature to produce turbidity. When passed over heated copper at $573\,\mathrm{K}$, it undergoes dehydration to produce an alkene of molecular formula $\mathrm{C_4H_8}$. The alcohol is:

(A) Butan-1-ol

(B) Butan-2-ol

(C) 2-Methylpropan-2-ol

(D) 2-Methylpropan-1-ol

Correct Answer: (C)

Step-by-step Explanation:

1. Instant turbidity with the Lucas reagent at room temperature confirms a **tertiary (3°) alcohol**.
2. Tertiary alcohols undergo elimination rather than dehydrogenation when passed over heated copper at $573\,\mathrm{K}$.
3. 2-Methylpropan-2-ol (tert-butyl alcohol, $(\mathrm{CH_3})_3\mathrm{C}-\mathrm{OH}$) is the only 3° isomer of $\mathrm{C_4H_{10}O}$. It reacts instantly with Lucas reagent and dehydrates over hot copper to give 2-methylpropene ($\mathrm{C_4H_8}$).

BOSONANCE

Practice Question 16 SINGLE-CORRECT MCQ

Ethanol is commercially converted into diethyl ether by heating with concentrated sulfuric acid at $413\,\mathrm{K}$. If the temperature is inadvertently raised to $443\,\mathrm{K}$, the predominant reaction product is:

(A) Ethene

(B) Ethyl hydrogen sulphate

(C) Diethyl sulphate

(D) Ethanal

Correct Answer: (A)

Step-by-step Explanation:

At $413\,\mathrm{K}$, bimolecular nucleophilic substitution ($\mathrm{S_N2}$) between two ethanol molecules forms ethoxyethane. At the higher temperature of $443\,\mathrm{K}$, intramolecular β-elimination ($\mathrm{E1}$) outcompetes substitution, and the protonated ethanol loses a proton to form **ethene** ($\mathrm{CH_2 = CH_2}$).

11. Common NEET Pitfalls & High-Yield Revision Checklist

NEET Exam Pitfall Traps

- **Alkene Hydration vs. Hydroboration-Oxidation:** Acid-catalyzed hydration ($\mathrm{H^+/H_2O}$) involves a carbocation intermediate and will rearrange (e.g., 3,3-dimethylbut-1-ene gives 2,3-dimethylbutan-2-ol). Hydroboration-oxidation gives the un-rearranged anti-Markovnikov product (3,3-dimethylbutan-1-ol).
- **Williamson Synthesis Pitfall:** Do not attempt to prepare a tertiary ether using a tertiary alkyl halide with an alkoxide. Tertiary halides undergo 100% E2 elimination with basic alkoxides to form alkenes. The halide must be 1°.
- **Reimer-Tiemann Intermediate:** The electrophile in the Reimer-Tiemann reaction is **dichlorocarbene (**: $\mathrm{CCl_2}$**)**, not the trichloromethyl carbanion or chloroform. It contains 6 valence electrons and is electrically neutral.
- **Ether Cleavage with Tertiary Groups:** Do not blindly apply the "iodide goes to smaller alkyl group" rule. If either alkyl group is tertiary (3°), the reaction proceeds via an $\mathrm{S_N1}$ **pathway**, and the iodide bonds to the *tertiary* carbon.
- **Alcohol Acidity vs. Water:** Water is more acidic than all aliphatic alcohols except methanol. Methanol is marginally more acidic than water.

Rapid Revision Checklist

- ☑ **Phenol Synthesis from Cumene:** Cumene + $\mathrm{O_2} \longrightarrow$ Cumene hydroperoxide $\xrightarrow{\mathrm{H^+}} \mathrm{Phenol + Acetone}$.
- ☑ **Bromination of Phenol:** $\mathrm{Br_2/CS_2} \longrightarrow$ p-Bromophenol; $\mathrm{Br_2/H_2O} \longrightarrow$ 2,4,6-Tribromophenol (white ppt).
- ☑ **Kolbe's vs. Reimer-Tiemann:** Kolbe introduces $-\mathrm{COOH}$ using $\mathrm{CO_2}$; Reimer-Tiemann introduces $-\mathrm{CHO}$ using $\mathrm{CHCl_3/NaOH}$.
- ☑ **Lucas Test:** 3° alcohols $\longrightarrow$ instant turbidity; $2^\circ \longrightarrow$ 5 minutes; $1^\circ \longrightarrow$ clear at room temperature.
- ☑ **Dehydrogenation over** Cu **at** $573\,\mathrm{K}$: $1^\circ \longrightarrow$ Aldehyde; $2^\circ \longrightarrow$ Ketone; $3^\circ \longrightarrow$ Alkene (dehydration).
- ☑ **Aspirin Synthesis:** Salicylic acid + Acetic anhydride $\xrightarrow{\mathrm{H^+}}$ Acetylsalicylic acid + Acetic acid.

12. Concise Summary Matrix

Table 6: Comprehensive Master Diagnostic Matrix for Alcohols, Phenols, and Ethers

Functional Class	Defining Structural Feature	Primary Synthetic Route	Distinguishing Chemical Test
Primary (1°) Alcohol	$-\mathrm{CH_2OH}$ group bound to sp^3 carbon	Hydroboration-oxidation of terminal alkenes; reduction of aldehydes	No turbidity with Lucas reagent at room temp; dehydrogenates to aldehyde over Cu at $573\,\mathrm{K}$.
Secondary (2°) Alcohol	$>\mathrm{CHOH}$ group bound to sp^3 carbon	Markovnikov hydration of alkenes; reduction of ketones	Turbidity with Lucas reagent within 5 minutes; dehydrogenates to ketone over Cu at $573\,\mathrm{K}$.
Tertiary (3°) Alcohol	$-\mathrm{C(OH)}<$ group bound to sp^3 carbon	Grignard addition to ketones followed by hydrolysis	Instantaneous turbidity with Lucas reagent; undergoes dehydration to alkene over Cu at $573\,\mathrm{K}$.
Phenol	$-\mathrm{OH}$ attached directly to sp^2 aromatic carbon	Aerial oxidation of cumene; acidification of sodium phenoxide	Violet coloration with neutral $\mathrm{FeCl_3}$; white precipitate with bromine water.
Ether	$\mathrm{R-O-R'}$ with sp^3 oxygen linkage	Williamson synthesis ($\mathrm{R'O^- + R-X}$, where $\mathrm{R-X}$ is 1°)	Cold concentrated HI cleavage yielding distinctive alkyl iodides.

BOSONANCE